Machine Learning for Economics & Finance

A Practical Introduction for Econ & Finance Majors

- Supervised Learning
- Neural Networks
- NLP & LLMs
- Forecasting
- Risk & Portfolio
- FinTech Applications

Sogang University · AI × Finance

Why Machine Learning for Finance & Economics?

Explosion of Data

Markets now generate terabytes daily — tick data, news, satellite imagery, social media. Traditional statistics can't scale.

Pattern Complexity

Nonlinear relationships between macro factors and asset prices defy classical linear models like OLS and VAR.

Predictive Power

ML models consistently outperform benchmark models in volatility forecasting, credit scoring, and fraud detection.

Automation & Scale

Algorithmic trading, robo-advisors, and real-time risk systems require models that learn and adapt without human intervention.

ML is not replacing economists — it is giving them a new, more powerful toolkit.

The ML Landscape: Three Core Paradigms

Supervised Learning

- Linear/Logistic Regression
- Decision Trees / XGBoost
- Neural Networks
- Support Vector Machines

Finance Use Case:

Credit scoring, stock return prediction, fraud detection

Unsupervised Learning

- K-Means Clustering
- PCA / t-SNE
- Autoencoders
- Gaussian Mixture Models

Finance Use Case:

Market regime detection, portfolio clustering, anomaly detection

Reinforcement Learning

- Q-Learning
- Deep Q-Networks (DQN)
- Policy Gradient
- Multi-Armed Bandit

Finance Use Case:

Optimal execution, dynamic portfolio rebalancing, market making

Most fintech applications today rely on Supervised Learning. RL is emerging in trading.

The ML Workflow: From Raw Data to Decisions

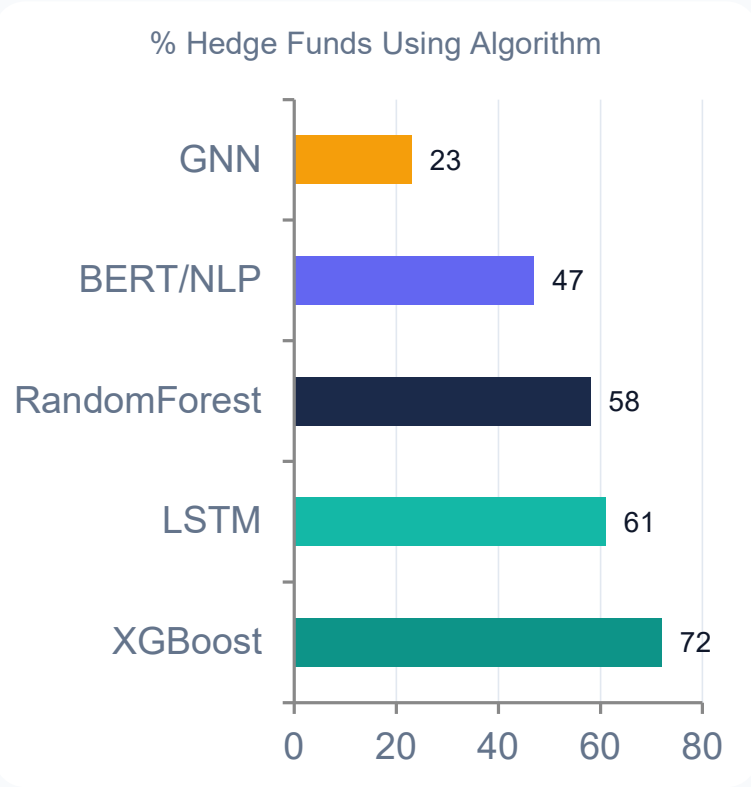
1 Data Collection	2 Feature Engineering	3 Model Training	4 Validation & Backtest	5 Deployment & Monitoring
Market prices, fundamentals, news, alt data	Returns, volatility, ratios, text embeddings	Select algorithm, tune hyperparameters	Walk-forward CV, Sharpe, drawdown metrics	Live inference, model drift detection

⚠ In Finance: Data leakage is the #1 mistake. Always use past data only to predict future. Walk-forward validation > k-fold CV.

The pipeline is iterative — bad features hurt more than a bad model choice.

Key Algorithms & Their Finance Applications

Algorithm	Finance Task	Strength
Linear Regression	Return forecasting	Interpretable, fast
Logistic Regression	Default prediction	Probabilities, auditable
Random Forest	Credit scoring	Robust, handles NaN
XGBoost / LightGBM	Stock ranking	State-of-art tabular data
LSTM / Transformer	Time-series prediction	Captures long dependencies
GNN	Sector spillover analysis	Models network structure
BERT / GPT	Sentiment analysis	Understands unstructured text



Source: Adapted from survey data on ML adoption in asset management (2022-2024 estimates).

Deep Learning: Neural Networks for Finance

How a Neural Network Learns

Input Layer:	Market features: price, volume, ratios, sentiment
Hidden Layers:	Learn nonlinear patterns automatically
Output Layer:	Predict: return, risk score, buy/sell signal
Backpropagation:	Adjusts weights to minimize prediction error
Gradient Descent:	Optimizer finds the best parameter values

CNN

Convolutional Neural Net

Chart pattern recognition from candlestick images

LSTM

Long Short-Term Memory

Sequential price & macro time-series forecasting

Transformer

Self-Attention Network

NLP sentiment, cross-asset correlation modeling

GNN

Graph Neural Network

Economic spillover, sector contagion analysis

In practice: LSTMs and Transformers are the most widely used in quantitative finance today.

NLP & Sentiment Analysis in Finance

Raw Text Sources

- FOMC meeting minutes
- Earnings call transcripts
- Financial news (Reuters, Bloomberg)
- Reddit / Twitter / StockTwits

▼

NLP Processing

- Tokenization & cleaning
- Named Entity Recognition (NER)
- Sentiment scoring (VADER, FinBERT)
- Topic modeling (LDA / BERTopic)

▼

Finance Signal

- Bull / Bear sentiment index
- Hawkish / Dovish Fed signal
- Earnings surprise predictor
- Market-moving event detector

FinBERT: Finance-Specific Language Model

- Pre-trained on 10-K filings, Reuters financial news
- Outperforms generic BERT on financial text
- Classifies news as: Positive / Negative / Neutral
- Powers hedge fund sentiment dashboards

Example: Fed Tone Detection

"The Committee seeks to achieve maximum employment and inflation at the rate of 2 percent."

FinBERT Score: Neutral (0.72) → Dovish Index: –0.12

NLP alpha: studies show central bank text sentiment predicts 10-year yield moves with significance.

ML for Time Series: Stock & Macro Forecasting

Classical vs ML Forecasting

ARIMA	✓ Interpretable, closed-form CI ✗ Linear only, manual order selection
VAR	✓ Multi-variable, Granger causality ✗ Curse of dimensionality
LSTM	✓ Nonlinear, long memory ✗ Black box, needs lots of data
Temporal Fusion Transformer	✓ SOTA on tabular time series, interpretable attention ✗ Complex to train, large compute
XGBoost (lag features)	✓ Fast, robust, works well in practice ✗ Doesn't capture temporal order natively

Walk-Forward Validation

Retrain on past window, test on next. Mimics real trading. Prevents leakage from future data.

Feature Engineering Tips

Log returns (not prices), rolling vol, RSI, macro surprises, earnings revision momentum.

Evaluation Metrics

For returns: IC (Information Coefficient), Sharpe. For classification: AUC-ROC, precision-recall. NOT just RMSE.

Key insight: predicting direction (up/down) is often more useful than predicting exact price level.

ML in Portfolio Management & Risk

📊 Portfolio Construction

- ML-based factor models (beyond Fama-French)
- Covariance estimation via RMT + ML
- Black-Litterman with ML views
- Regime-conditional allocation (HMM + RL)

🛡️ Risk Modeling

- Value-at-Risk via quantile regression forests
- Stress testing with GAN-generated scenarios
- Tail risk estimation (Extreme Value Theory + NN)
- Systemic risk: GNN contagion models

🔍 Factor Discovery

- LASSO / Elastic Net for factor selection
- Deep learning alpha factors from price data
- NLP factors from corporate disclosures
- Alternative data: satellite, credit card, ESG

⚡ Execution & Trading

- Optimal execution: Almgren-Chriss + RL
- Market impact prediction with LSTM
- Order book dynamics modeling
- High-frequency pattern detection with CNN

Modern portfolio management is increasingly a hybrid of classical MPT and ML-driven signal generation.

LLMs & Generative AI: The New Frontier

$900B+	78%	40%	10x
AI in finance market by 2030	Banks exploring GenAI in 2024	Reduction in research time (JPMorgan pilot)	Analyst productivity gain (estimates)

Research & Analysis

- Automated earnings report summaries
- 10-K/10-Q risk factor extraction
- Macro narrative generation from data
- Automated analyst note drafting

Chatbots & Advisory

- Robo-advisor natural language interface
- Portfolio explanation in plain English
- Customer query resolution (wealth mgmt)
- Regulatory Q&A (RAG over compliance docs)

Code & Model Generation

- Auto-generate backtesting code
- SQL query generation for financial DBs
- Model documentation automation
- Formula generation (Bloomberg, Excel)

Caution: LLMs hallucinate. Always validate outputs with ground truth data sources before trading decisions.

Pitfalls, Biases & Ethical Considerations

⚠ Common ML Mistakes in Finance

Data Leakage

Using future information in training. E.g., using next-day close in features.

Overfitting

Model memorizes train set. 1000 features, 200 samples = guaranteed overfit.

Survivorship Bias

Training only on stocks that still exist ignores delisted losers.

Look-Ahead Bias

Fundamental data (earnings) has reporting lag — use 'as-of' databases.

Non-Stationarity

Financial time series regimes change. Yesterday's model may fail tomorrow.

p-Hacking

Testing 1000 strategies finds false positives. Multiple testing correction required.

Fairness & Bias

ML credit models can perpetuate racial / gender bias from historical loan data. Disparate impact testing is required in many jurisdictions.

Explainability (XAI)

EU AI Act and Basel require model interpretability. SHAP, LIME, and attention maps help open the 'black box' for regulators.

Systemic Risk

If all funds use similar ML signals, they may trade simultaneously, amplifying crashes. Flash Crash 2010 as a warning.

A good quant is 90% statistician, 10% programmer — skepticism about results is a virtue, not a weakness.

Getting Started: Python for ML in Finance

Data Layer	pandas, numpy, yfinance, wrds, OpenBB
ML Framework	scikit-learn, XGBoost, LightGBM, CatBoost
Deep Learning	PyTorch, TensorFlow/Keras, Hugging Face
Backtesting	Backtrader, vectorbt, zipline-reloaded
Visualization	matplotlib, plotly, seaborn, streamlit

```python
# Predict next-day return with XGBoost

import yfinance as yf, xgboost as xgb

import pandas as pd, numpy as np


df = yf.download('AAPL', start='2020-01-01')

df['ret'] = df['Close'].pct_change()

df['lag1'] = df['ret'].shift(1)

df['vol'] = df['ret'].rolling(20).std()

df.dropna(inplace=True)


X = df[['lag1', 'vol']]

y = (df['ret'] > 0).astype(int)


model = xgb.XGBClassifier()

model.fit(X[:-60], y[:-60])

preds = model.predict(X[-60:])
```

Tip: Start with yfinance + scikit-learn. Graduate to PyTorch when you need custom architectures.

Career Paths: Where ML Meets Finance

Quantitative Analyst (Quant) — $150K–$500K+

Hedge Funds, Investment Banks

- Python/R, statistics, ML
- Signal research, backtesting
- Derivatives pricing

ML / AI Engineer (Finance) — $130K–$350K

FinTech, Banks, Crypto

- Deep learning, MLOps
- Real-time systems, Kafka
- Model deployment, AWS

Risk Data Scientist — $100K–$220K

Banks, Regulators, Insurance

- XGBoost, survival models
- Explainable AI (SHAP)
- Regulatory stress testing

FinTech Product / Data Analyst — $80K–$180K

Startups, Payments, WealthTech

- SQL, Python, dashboards
- A/B testing, funnel analysis
- NLP for customer insights

Salary ranges are USD estimates for US market (2024). Korea KRW equivalents vary; Tier-1 quant roles at domestic asset managers: ~~₩~~100M–~~₩~~300M+.

Key Takeaways

01	ML extends — not replaces — classical econometrics. Both toolkits are essential.
02	Data quality > model complexity. Clean, leakage-free features beat fancy algorithms.
03	Understand your evaluation metrics. Sharpe and IC matter more than RMSE in finance.
04	Interpretability matters for regulators, clients, and your own understanding.
05	LLMs are transforming financial research workflows — start exploring them now.